

J-space Phase 4.4 — terminal development synthesis

Operator convergence without a stable sparse causal instrument

Qwen A1000 decision, side-methods imports, and durability boundary — 2026-08-03; Terminal C closeout appended
2026-08-05 at jspace-phase4-frozen-v1

Scope. This is a living **development only** report. No Phase 4 confirmatory or replication intervention outcome was opened. The mechanical result is Q-L4: no single canonical sparse Qwen lens. This handout is a freeze-review input, not a review signature, PI approval, freeze commit, or publication claim.

1 Decision in one page

The sealed A1000 queue separated averaged-operator convergence from sparse-selection and causal-endpoint invariance. The former passes strongly; the latter do not.

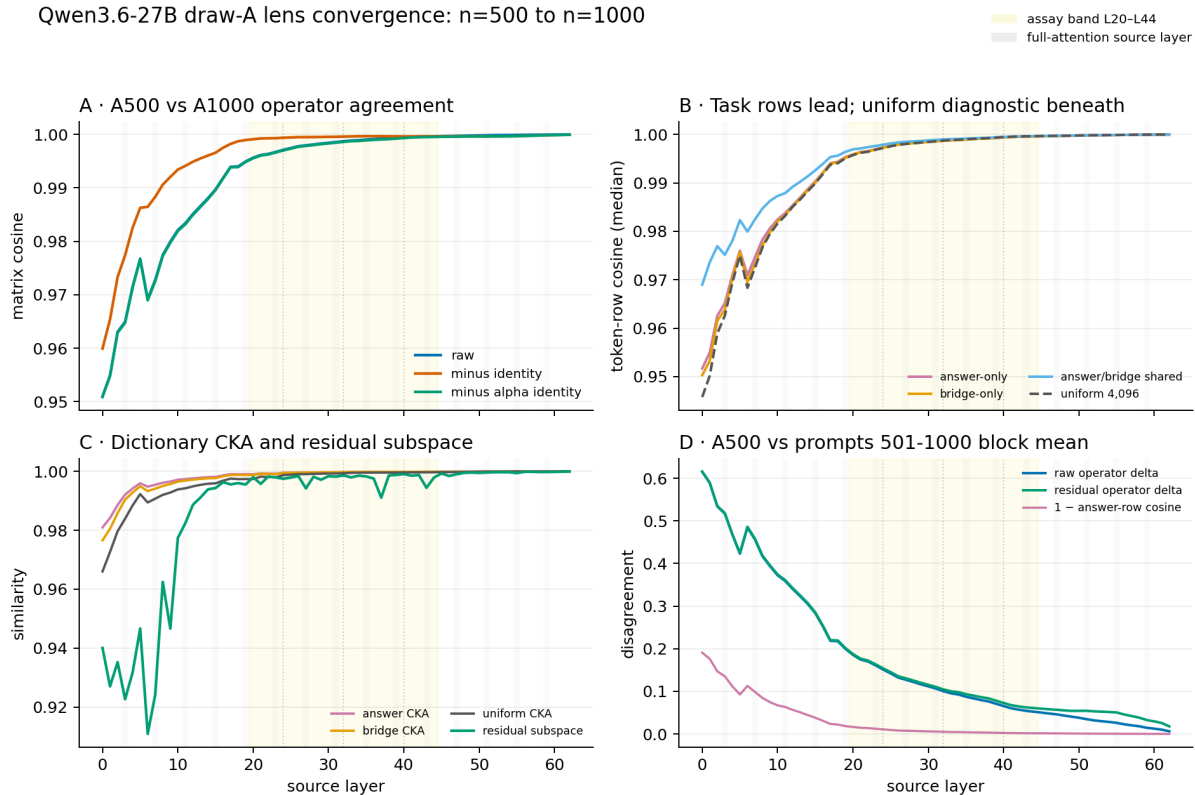
Gate	observed	threshold	disposition
structural task q50	0.998702	≥ 0.95	PASS
structural task q05	0.998122	≥ 0.90	PASS
selected-ID Jaccard	0.538462	frozen floor	FAIL
normalized projector overlap	0.709818	frozen floor	FAIL
bridge-rescue A500–A1000 difference	−0.294028 nat	frozen bound	FAIL
prompt-323 influence materiality	negligible	all frozen rulers	PASS

The all-position audit retains every one of 17,381 functional positions: 15,536 near-tie, 1,845 stable-core, and zero rank-deficient. Even stable-core median ID Jaccard is only 0.666667. Tied rows are therefore part, but not all, of the instability.

The unchanged Q-L table emits **Q-L4**. It nominates no sparse lens, blocks P4-P2, makes the Bank-B orthogonal shot inapplicable, and prevents M3 and M4 from opening. There is no A2000 route.

2 A1000 structural and functional evidence

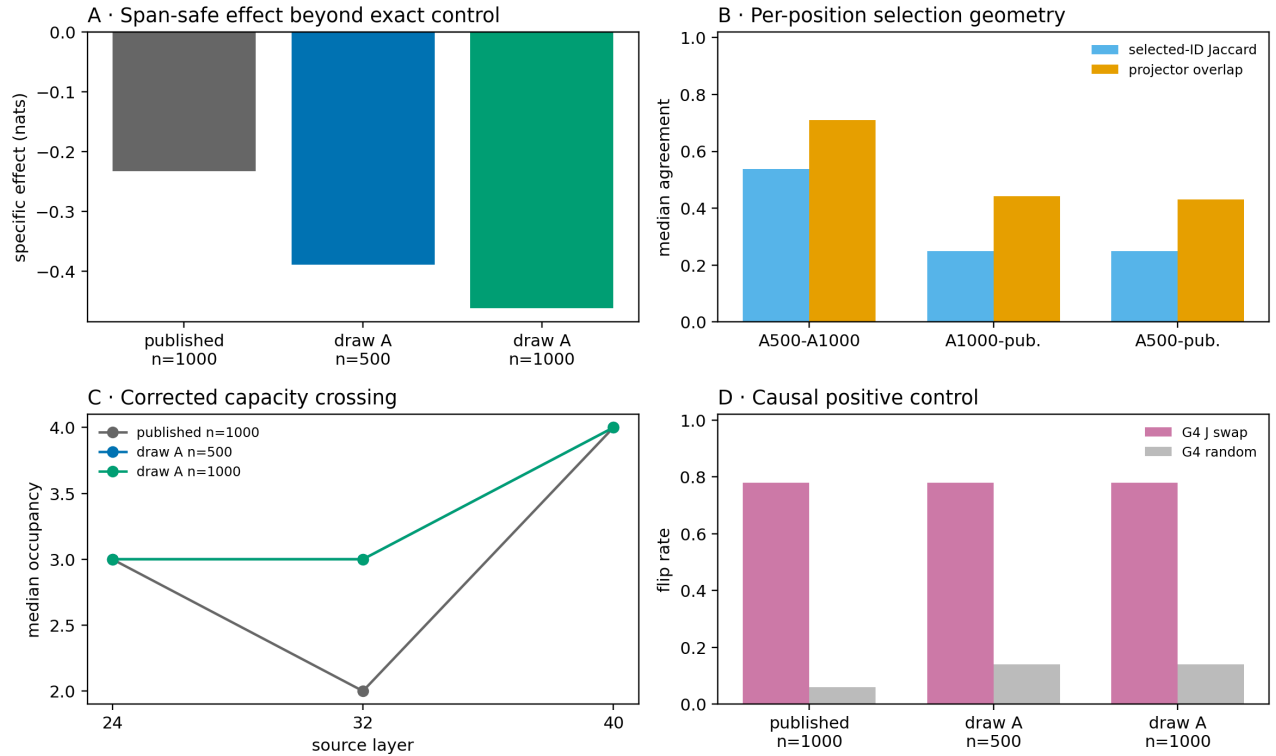
The A1000 artifact is the exact cumulative draw-A fit at 1,000 prompts. Its model snapshot, external published lens, 63-layer tensors, checkpoint mean, and registered hashes pass the fresh-VM integrity audit.



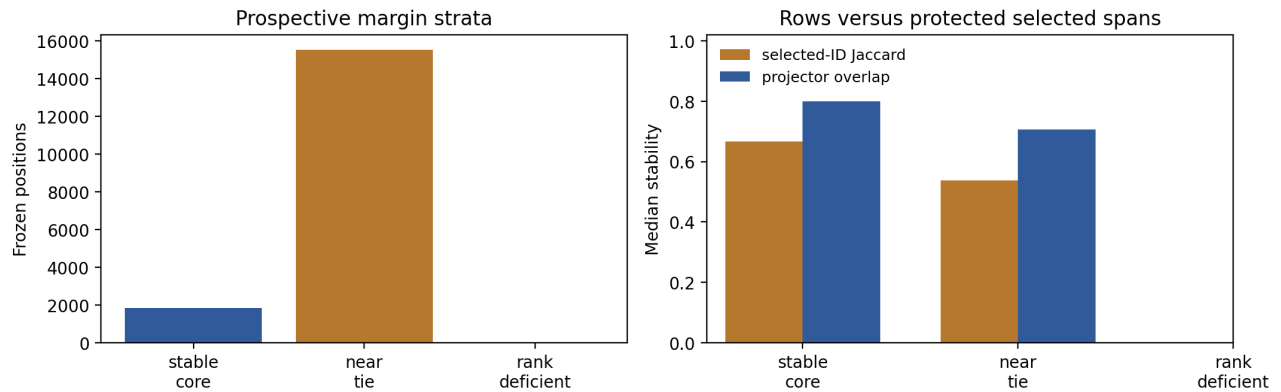
Development structural sensitivity. Published comparisons in the registered table use an external published reference, partially specified recipe.

High matrix similarity does not carry through to the discrete selection or every causal endpoint. Occupancy, corrected capacity, span-safe specificity, tail rate, G4, and bridge preference pass; selected IDs, selected projector, and bridge rescue fail.

Qwen A500-A1000 functional gate (Phase 4 development)
Published comparator: external published reference, partially specified recipe



Qwen A500-A1000 selection-margin audit



Development mechanism audit; every stratum remains in the functional decision.

Across the last three fit-size comparisons, selected-ID Jaccard repeats at approximately 0.53846. Projector overlap rises only from 0.67479 to 0.70302 to 0.709818. This repeated pattern is treated as a methods result, not as a reason to relax a threshold.

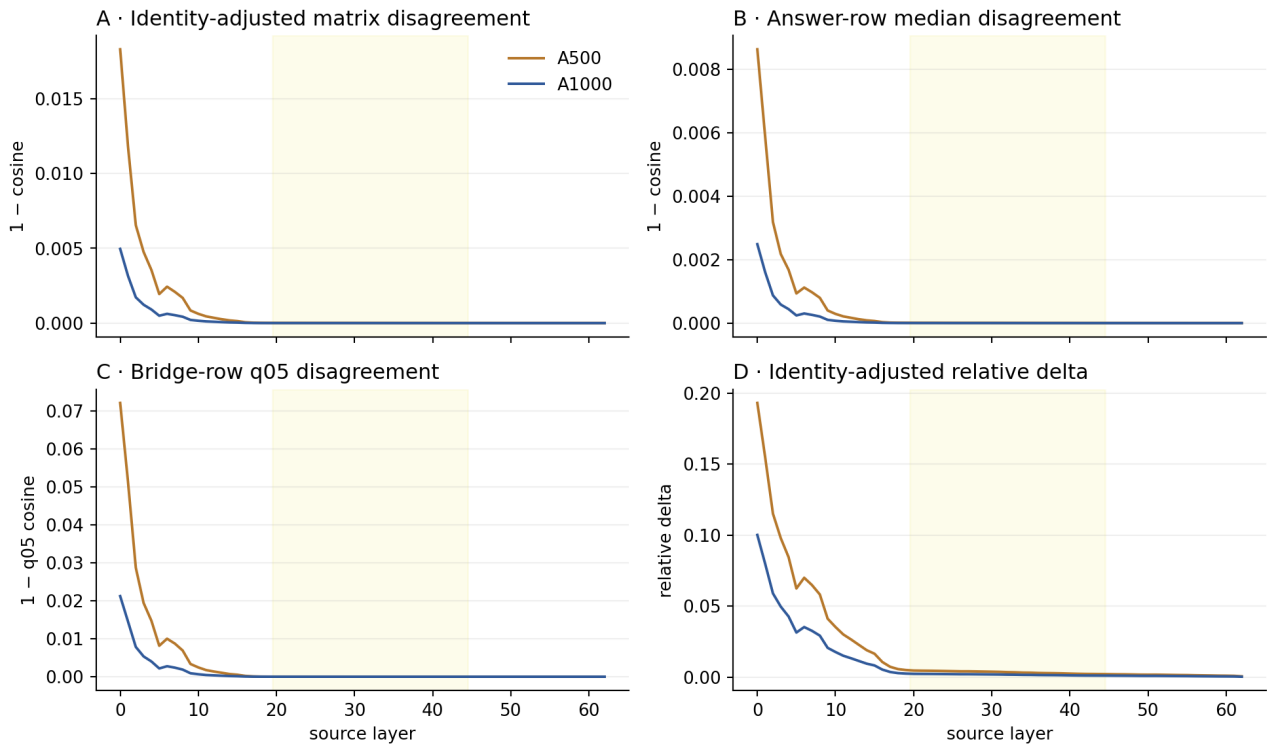
3 Retained-prompt influence and runtime boundary

The first prompt-323 attempt stopped before evidence because the current Jacobian norm disagreed with the historical fit-log control. Current repeats were stable; the prompt-112 control showed that the mismatch was broader than one prompt. Nominal versions and model bytes matched, but historical distribution-content and compiled-kernel identities were unavailable.

Because retained-prompt influence cannot alter Q-L1–Q-L5, a prospective amendment fixed exact current distribution contents, a primary computation, a discarded repeatability computation, and the unchanged 0.5 within-runtime norm tolerance before any new influence output. It changed no bank, model, lens, endpoint, sample, materiality threshold, or SESOI.

Primary and repeat maxima are 181.776618 and 181.777423; the worst normalized per-layer difference is 0.004572. All tensors are finite. The A500 and A1000 materiality values are tiny, with the closest more than 3,800-fold below its frozen threshold. Influence peaks early and is essentially zero through assay layers 20–44.

Qwen draw-A prompt-323 retained-influence audit



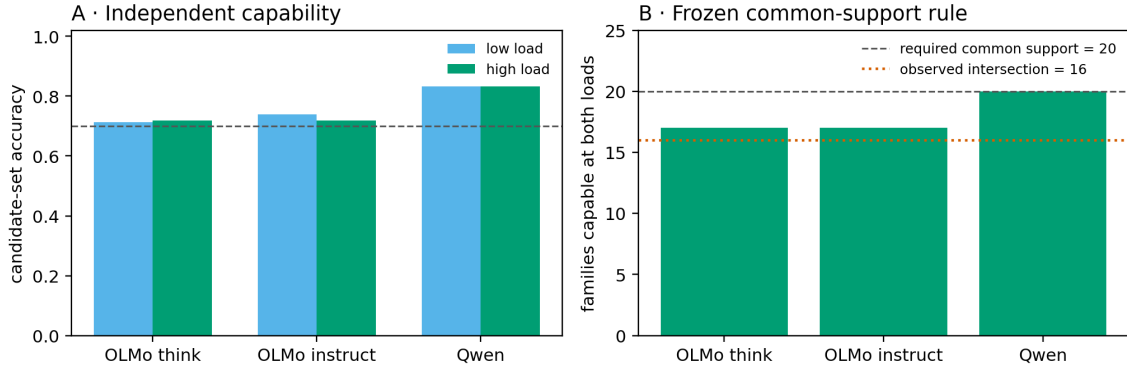
Exact equal-prompt algebra on A500 and A1000; prompt 323 is never removed from a registered lens.

This licenses only an exact-pinned *current-runtime sensitivity shape*. It does not establish historical-runtime reproducibility.

4 Primary-family disposition

Endpoint	terminal state	reason
P4-P1	estimation-only	Bank B cannot power the 0.25-nat joint target; Q-L4 makes the conditional orthogonal shot not applicable.
P4-P2	blocked / removed	Its legal route requires Q-L1/Q-L2 and external producer review; no pilot, power run, or intervention opened.
P4-P3	blocked	Strict OLMo import and fresh mainline replay reproduce 16 common-capable families against 20 required.

No primary entered the Holm family and no retired endpoint donated alpha. Bank W’s load intervention is not lawfully runnable as a cross-model Phase 4 primary at current capability support; it routes to Phase 5B as per-model estimation.



5 Side releases: admit, then cite narrowly

- **Gemma transport:** methods blocker at side-development-import tier. Selected-slot tangents are bit-identical, but the frozen all-slot relative-error criterion fails at 0.002458 versus 10^{-5} . No mechanism, nondifferentiability, workspace, or intervention claim follows.
- **OLMo lineage:** methods/development import only. O1 preserves the 16/20 block; O2 is consistent with broadly conserved capacity recruitment; O3 is a dictionary-formation pattern; O5 has no identifiable estimand.
- No native side-study evidence ID was inserted into the Phase 4 registry, and neither import upgrades its source tier.

The OLMo trajectory continues to support development localization across own and common frames, but it does not identify external-state substitution or a cross-model causal mechanism.

6 Durability and release boundary

Exact A120 capacity recovery reproduces the old registered SHA-256. The historical A120–A250 operational state remains irrecoverable and was never synthesized. An append-only methods event classifies that source as partially durable, with its decision role superseded by later live A250–A500 and A500–A1000 functional gates.

Two same-mounted audits agree on:

registry rows	78
live events	61
live output references	419
verified references	418
known deficits	1
unexpected deficits / pin conflicts	0 / 0

The second audit is a same-mount repeat, not an independent Drive rematerialization. The pre-freeze inventory therefore remains honestly `NOT_REVIEW_READY` for freeze despite passing ancestry, policy, path, native-ID, and temporary-leak checks.

7 Terminal interpretation

Phase 4 ends at Terminal C and paper path B/C: a development/methods instrument study with no confirmatory primary. Its central useful result is negative but specific: averaged transport can converge while sparse selection and a load-bearing causal endpoint remain fit-sensitive. Negligible retained-prompt influence does not rescue the instrument.

The next actions are external: fresh Drive rematerialization, independent review of the Q-L4/runtime/untouched-data and archival boundaries, and PI disposition. No freeze commit or tag is authorized by this handout.

8 Terminal C closeout (appended 2026-08-05, post-freeze)

The external actions above completed, and Phase 4 froze at `jSPACE-phase4-frozen-v1` (commit `ea63522`; full hash in `FREEZE_HANDOFF.md`):

- **Study-2 import boundary.** `p4-import-gemma-transport-study2-v1`: the unchanged five-layer `local_tangent_mismatch` classifier is a closed exact-JVP finite-scale methods result under the target-blind pooled ceiling 0.07870368901355948 (historical all-slot error 0.002458111383020878 within it; selected slot bit-identical). `p4-import-olmo-lineage-study2-v1`: the official SFT/DPO wedge is capability-gated (972-row batteries, 0.617%/0.309% capable, zero Bank-S facts on direct+composed; effects missing, not zero); H6 licenses no in-band finite-dose regime at L24/L32/L40 (Think-only L56 $\varepsilon=0.10$ late anchor); registered-dose coverage is archive-unavailable, not 0%; the Bank-W pair power is 0.7788 at 16 shared capable families (first passing count 18) — a planning closure. Native tiers preserved; no native side-study ID entered the Phase 4 registry.
- **Runtime-identity methods object** (`p4-runtime-identity-synthesis-v1`): fit-era and current-era backward semantics are not interchangeable under version-level pinning alone; the prompt-323 influence shape is licensed for the exact-pinned current runtime only.
- **Durability at freeze.** Fresh VM/mount/clone materialization: 521 live output references, 520 verified (515 literal, 3 repository-materialization, 2 append-only-prefix), one failure = the accepted permanent A120–A250 `state.json` deficit, zero unexpected, zero pin conflicts. Whole-registry durability is honestly 520/521; “all artifacts verify” is never claimed.
- **Review and disposition.** Narrative-blind independent review 13/14 PASS + 1 PASS-with-limitation (errata fixed); untouched-data audit PASS; PI disposition accepted 13/13; freeze authorized and tagged. Suites at freeze: phase4 302, gemma 71, olmo 90.

The terminal sentence of record: *Phase 4 is frozen at Q-L4 Terminal C as a development/methods release; its main contribution is the demonstrated separation among averaged-operator convergence, sparse-coordinate invariance, and causal-endpoint invariance, and no Phase 4 confirmatory primary or intervention outcome was opened.* Post-freeze analysis lives on `interp-jSPACE_paper_analysis` under `jSPACE_paper/analysis/`; this handout is superseded for claims by `PHASE4_STATE_OF_RECORD.md` and the release claim ledger.